

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0510

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 10%

Code of Conduct:

I U. Udhayashree-192521448 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: U.Udhayashree

Rubrics for Evaluation

<i>Criteria</i>	<i>Weightage</i>	<i>Level 4 – Exemplary</i>	<i>Level 3 – Proficient</i>	<i>Level 2 – Developing</i>	<i>Level 1 – Beginning</i>	<i>Marks(100)</i>
<i>Understanding of Concepts</i>	<i>25%</i>	<i>Demonstrates thorough, accurate understanding of concepts</i>	<i>Good understanding with minor gaps</i>	<i>Basic understanding with some misconceptions</i>	<i>Poor understanding; major misconceptions</i>	
<i>Explanation</i>	<i>25%</i>	<i>Detailed, well-explained answers with relevant examples</i>	<i>Clear explanation with adequate details</i>	<i>Limited explanation; lacks depth</i>	<i>Poor or unclear explanation</i>	
<i>Application</i>	<i>20%</i>	<i>Effectively applies concepts with strong analytical ability</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>	
<i>Organization</i>	<i>15%</i>	<i>Well-structured answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization, incoherent answers</i>	

<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>	
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>	

1. An image appears too bright.

Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.

Answer:

A **Negative Transformation** is the most suitable grey level transformation for an image that appears too bright. It reverses the intensity values, converting bright pixels into dark pixels and dark pixels into bright pixels. This helps reveal details that may be hidden in over-bright regions.

Formula:

$$s = L - 1 - r$$

Where:

- r = Input pixel intensity
- s = Output pixel intensity
- L = Total number of grey levels (256 for an 8-bit image)

Concept:

Negative transformation inverts the grey levels of an image. High-intensity pixels become low-intensity pixels, improving the visibility of details in very bright areas.

Application Process:

1. Read the input image.
2. Apply the negative transformation formula to every pixel.
3. Generate the transformed image with inverted grey levels.
4. Display the enhanced image.

Expected Outcome:

- Reduces excessive brightness.
- Improves visibility of hidden details.
- Enhances overall image contrast.
- Produces a clearer image for further processing.

Justification:

Negative transformation is suitable for overly bright images because it effectively suppresses high-intensity regions and enhances features that are difficult to observe in the original image. It is widely used in image enhancement and medical image processing to improve visual interpretation.

2.A dark image lacks detail.

Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.

Answer:

A **Log Transformation** is the most suitable grey level transformation for a dark image because it enhances **low-intensity (dark) pixels** while compressing **high-intensity (bright) pixels**. This makes hidden details in dark regions more visible without significantly affecting the brighter areas.

Formula:

$$y = c \log(1+x)$$

Where:

- r = Input pixel intensity
- s = Output pixel intensity
- c = Constant used for scaling

Concept:

Log transformation is a grey level transformation that expands the values of low-intensity pixels and compresses the values of high-intensity pixels. As a result, dark areas become brighter and more visible.

Application Process:

1. Read the input image.
2. Apply the log transformation to every pixel using the formula.

3. Scale the transformed values to the range 0–255.
4. Display the enhanced image.

Expected Outcome:

- Enhances dark regions.
- Reveals hidden details in low-intensity areas.
- Improves overall image visibility.
- Produces a brighter and clearer image.

Justification:

Log transformation is ideal for dark images because it gives greater emphasis to low-intensity pixels, making details in shadowed or underexposed regions easier to observe. It is widely used in image enhancement, medical imaging, and satellite image processing to improve visibility without excessively increasing the brightness of already bright regions.

3. An image has poor contrast.

Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.

Answer:

The most suitable histogram-based technique for improving image contrast is **Histogram Equalization (HE)**. It redistributes the pixel intensity values so that they are spread more uniformly over the available grey level range, resulting in better contrast and improved image quality.

Concept:

Histogram Equalization is an image enhancement technique that increases the global contrast of an image by transforming the intensity distribution. It is especially effective when the image has poor contrast due to most pixel values being concentrated in a narrow intensity range.

Application Process:

1. Read the input image.
2. Calculate the histogram of pixel intensities.
3. Compute the **Cumulative Distribution Function (CDF)**.
4. Map the original pixel values to new intensity values using the CDF.
5. Generate the enhanced image with improved contrast.

Working Formula:

$$s = (L - 1) \times CDF(r)$$

Where:

- r = Original pixel intensity
- s = Transformed pixel intensity
- L = Number of grey levels (256 for an 8-bit image)
- **CDF** = Cumulative Distribution Function

Expected Outcome:

- Improves overall image contrast.
- Enhances visibility of objects and fine details.
- Spreads intensity values across the full grey level range.
- Produces a clearer and more visually appealing image.

Justification:

Histogram Equalization is the best choice for poor-contrast images because it automatically redistributes pixel intensities over the entire intensity range. This increases the distinction between dark and bright regions, making important features easier to identify and improving the image for further analysis.

4.A low-contrast image requires uniform intensity distribution.

Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.

Answer:

The most suitable method is **Histogram Equalization (HE)**. It is used to achieve a **uniform intensity distribution** by spreading the pixel values across the entire available grey level range. This improves the contrast of low-contrast images and makes image details more visible.

Concept:

Histogram Equalization is an image enhancement technique that redistributes the intensity values of an image so that the histogram becomes more uniform. This increases the dynamic range of pixel intensities and enhances the overall contrast.

Working Process:

1. Read the input image.
2. Compute the histogram of pixel intensity values.
3. Calculate the **Cumulative Distribution Function (CDF)**.
4. Transform each pixel intensity using the CDF.
5. Generate the enhanced image with a more uniform intensity distribution.

Transformation Formula:

$$s = (L - 1) \times CDF(r)$$

Where:

- r = Original pixel intensity
- s = New pixel intensity
- L = Number of grey levels (256 for an 8-bit image)
- **CDF** = Cumulative Distribution Function

Effect on Intensity Levels:

- Spreads pixel intensities uniformly over the full grey level range.
- Increases image contrast.
- Enhances visibility of dark and bright regions.
- Makes edges and fine details more distinguishable.

Justification:

Histogram Equalization is the best choice for a low-contrast image because it redistributes intensity values uniformly, improving the contrast without changing the image content. It enhances the visibility of important features and produces a clearer image for analysis and further image processing.

5. An image shows uneven brightness across regions.

Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.

Answer:

The most suitable enhancement technique for correcting **non-uniform illumination** is **Adaptive Histogram Equalization (AHE)** or **Contrast Limited Adaptive Histogram Equalization (CLAHE)**. CLAHE is preferred because it improves local contrast while preventing excessive noise amplification.

Concept:

CLAHE divides the image into small regions (tiles) and applies histogram equalization to each region separately. This enhances local contrast and corrects uneven brightness across different parts of the image.

Working Process:

1. Read the input image.
2. Divide the image into small tiles.
3. Apply histogram equalization to each tile independently.
4. Limit the contrast to avoid over-enhancement (Contrast Limiting).
5. Merge all tiles using interpolation to produce the final enhanced image.

Expected Outcome:

- Corrects uneven brightness.
- Enhances local contrast in different regions.
- Improves visibility of hidden details.
- Produces a natural-looking image with balanced illumination.

Justification:

CLAHE is the most appropriate technique because it enhances each region independently instead of processing the whole image at once. This effectively corrects non-uniform illumination, preserves image details, and avoids over-enhancement or noise, making it suitable for medical images, satellite images, and low-light photographs.

6.A noisy image contains random intensity variations.

Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

Answer:

The most suitable smoothing filter for removing random intensity variations is the **Median Filter**. It is especially effective for removing **salt-and-pepper noise** while preserving image edges.

Concept:

A Median Filter is a **non-linear spatial filter** that replaces the center pixel with the **median value** of the neighboring pixels instead of the average. This removes noise without significantly blurring edges.

Working Mechanism:

1. Read the noisy image.
2. Select a neighborhood window (e.g., 3×3 or 5×5).
3. Arrange the pixel values in ascending order.
4. Find the median value.
5. Replace the center pixel with the median value.
6. Repeat the process for all pixels in the image.

Impact on Noise Reduction:

- Removes random **salt-and-pepper noise** effectively.
- Preserves edges better than averaging filters.
- Produces a smoother and cleaner image.
- Improves image quality for further processing.

Justification:

The Median Filter is the best choice because it eliminates random intensity variations without significantly affecting image details or edges. Unlike mean filters, it does not blur sharp boundaries, making it widely used in image enhancement and preprocessing.

7. An image requires noise reduction without significant loss of details.

Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.

Answer:

The most suitable filtering approach is the **Bilateral Filter**. It reduces image noise while preserving important edges and fine details, making it ideal when detail preservation is required.

Concept:

The Bilateral Filter is a **non-linear smoothing filter** that considers both the **spatial distance** and the **intensity difference** between neighboring pixels. It smooths similar pixels while preserving edges.

Working Process:

1. Read the noisy image.
2. Select a neighborhood around each pixel.
3. Compute weights based on:
 - Spatial distance between pixels.
 - Intensity difference between pixels.
4. Calculate the weighted average of neighboring pixels.
5. Replace the center pixel with the computed value.
6. Repeat the process for all pixels.

Expected Outcome:

- Reduces image noise effectively.
- Preserves edges and fine details.
- Produces a smooth and natural-looking image.
- Improves image quality for further processing.

Justification:

The Bilateral Filter is the best choice because it balances **noise removal** and **detail preservation**. Unlike Gaussian or Mean filters, it smooths only similar pixels while protecting edge information. This makes it widely used in medical imaging, computer vision, and photographic image enhancement.

8. An image appears blurred after smoothing.

Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

Answer:

The most suitable sharpening method is **Laplacian Sharpening**. It enhances image details by emphasizing high-frequency components such as edges and fine structures, making the blurred image appear sharper.

Concept:

Laplacian Sharpening is a **second-order derivative** technique used to detect edges. It highlights regions with rapid intensity changes and adds these details back to the original image to improve sharpness.

Working Principle:

1. Read the blurred image.
2. Apply the **Laplacian operator** to detect edges.
3. Subtract the Laplacian image from the original image.
4. Generate the sharpened image with enhanced details.

Formula:

$$g(x, y) = f(x, y) - \nabla^2 f(x, y)$$

Where:

- $f(x, y)$ = Original image
- $\nabla^2 f(x, y)$ = Laplacian of the image
- $g(x, y)$ = Sharpened image

Expected Results:

- Restores blurred edges.
- Enhances fine details and textures.
- Improves overall image sharpness.
- Produces a clearer image for further analysis.

Justification:

Laplacian Sharpening is effective because it enhances **high-frequency components** that are lost during smoothing. It restores edge information without significantly affecting the overall brightness, making it widely used in image enhancement and computer vision applications.

9. Fine details in an image are not clearly visible.

Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

Answer:

The most suitable filtering technique is the **High-Pass Filter (HPF)**. It enhances fine details by emphasizing the **high-frequency components** of an image, such as edges, lines, and textures.

Concept:

A High-Pass Filter allows **high-frequency components** to pass while suppressing **low-frequency components**. Since fine details and edges correspond to high frequencies, the filter makes them more prominent and improves image sharpness.

Working Process:

1. Read the input image.
2. Apply a High-Pass Filter kernel to the image.
3. The filter suppresses low-frequency (smooth) regions.
4. High-frequency components (edges and fine details) are enhanced.
5. Generate the sharpened output image.

Effect on High-Frequency Components:

- Enhances edges and object boundaries.
- Improves visibility of fine details and textures.
- Increases image sharpness.
- Removes smooth background information while highlighting important features.

Justification:

The High-Pass Filter is the most suitable technique because it emphasizes high-frequency information responsible for image details. By suppressing low-frequency components and enhancing edges, it restores clarity and makes fine features easier to identify. It is widely used in image enhancement, edge detection, and computer vision applications.

10. A noisy image must be preprocessed before further analysis.

Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

Answer:

The most suitable spatial filtering method is the **Gaussian Filter**. It is widely used in image preprocessing to reduce **Gaussian noise** while preserving the overall structure of the image.

Concept:

A Gaussian Filter is a **linear spatial smoothing filter** that uses a Gaussian kernel to perform a weighted average of neighboring pixels. Pixels closer to the center have higher weights than distant pixels, resulting in effective noise reduction.

Working Process:

1. Read the noisy input image.
2. Select an appropriate Gaussian kernel (e.g., **3 × 3** or **5 × 5**).
3. Convolve the Gaussian kernel with the image.
4. Replace each pixel with the weighted average of its neighboring pixels.
5. Generate the smoothed image for further processing.

Gaussian Formula:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

Where:

- $G(x, y)$ = Gaussian function
- x, y = Pixel coordinates
- σ = Standard deviation controlling the amount of smoothing

Expected Outcome:

- Reduces Gaussian noise effectively.
- Produces a smooth and clean image.
- Preserves the overall image structure.

- Improves the quality of the image for segmentation, feature extraction, and object detection.

Justification:

The Gaussian Filter is the preferred preprocessing technique because it effectively removes random noise while maintaining the overall appearance of the image. It provides a smooth output without introducing significant artifacts, making it an essential step before performing further image analysis or computer vision tasks.